AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Csquare SFO2, CA -- station KSJC, America/Los_Angeles
Committed pair	OSM 54119665 -> 54121151 Csquare SFO2 / QTS Santa Clara 1
Facade gap	339.2 m between the two halls
Weather record	43,747 real hourly records from KSJC
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.1944 C at 325 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2023-07-21 (knife_edge)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 2 mode changes. 1 hour was safe but still ran chillers, because the schedule could not afford them.

Agent	8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	1 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	16.255	21.0	16.264	0.907
01	FREE	yes	-	16.562	21.0	16.684	0.848
02	FREE	yes	-	16.699	21.0	16.731	0.824
03	FREE	yes	-	16.699	21.0	16.731	0.738
04	FREE	yes	-	16.974	21.0	16.171	0.948
05	FREE	yes	-	16.974	21.0	16.171	0.886
06	FREE	yes	-	17.259	21.0	16.731	0.689
07	FREE	yes	-	18.760	21.0	19.027	0.874
08	mechanical	no	dew point	20.589	21.0	20.061	1.191

09	mechanical	no	dry-bulb	23.644	21.0	22.841	1.736
10	mechanical	no	dry-bulb	26.191	21.0	24.594	1.798
11	mechanical	no	dry-bulb	27.523	21.0	26.853	1.503
12	mechanical	no	dry-bulb	29.481	21.0	28.385	1.547
13	mechanical	no	dry-bulb	30.193	21.0	29.598	1.481
14	mechanical	no	dry-bulb	30.743	21.0	30.158	1.348
15	mechanical	no	dry-bulb	30.679	21.0	30.155	1.194
16	mechanical	no	dry-bulb	29.959	21.0	29.595	1.003
17	mechanical	no	dry-bulb	29.411	21.0	29.045	1.184
18	mechanical	no	dry-bulb	27.977	21.0	27.378	1.135
19	mechanical	no	dry-bulb	26.536	21.0	24.598	1.395
20	mechanical	no	dry-bulb	24.606	21.0	21.268	1.484
21	mechanical	no	dry-bulb	23.840	21.0	20.718	1.491
22	mechanical	no	dry-bulb	21.513	21.0	18.488	1.071
23	mechanical	yes	switch budget	19.560	21.0	17.934	0.814

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.255 C, which is 4.745 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.907 C, and the margin is measured, not chosen: 0.762 C of group-conditional forecast error for this hour of day, plus 0.1452 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.562 C, which is 4.438 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.848 C, and the margin is measured, not chosen: 0.676 C of group-conditional forecast error for this hour of day, plus 0.1722 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.699 C, which is 4.301 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.824 C, and the margin is measured, not chosen: 0.577 C of group-conditional forecast error for this hour of day, plus 0.2476 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.699 C, which is 4.301 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.738 C, and the margin is measured, not chosen: 0.490 C of group-conditional forecast error for this hour of day, plus 0.2476 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.974 C, which is 4.026 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.948 C, and the margin is measured, not chosen: 0.701 C of group-conditional forecast error for this hour of day, plus 0.2476 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.974 C, which is 4.026 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.886 C, and the margin is measured, not chosen: 0.639 C of group-conditional forecast error for this hour of day, plus 0.2476 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.259 C, which is 3.741 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.689 C, and the margin is measured, not chosen: 0.441 C of group-conditional forecast error for this hour of day, plus 0.2476 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.760 C, which is 2.240 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.874 C, and the margin is measured, not chosen: 0.723 C of group-conditional forecast error for this hour of day, plus 0.1502 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 20.589 C would have passed. The outside air is too HUMID: the dew-point bound is 15.01 C against a 15.0 C maximum, failing by 0.01 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.644 C against a 21.0 C limit, so it fails by 2.644 C. It would take a limit 2.644 C higher, or a bound 2.644 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.191 C against a 21.0 C limit, so it fails by 5.191 C. It would take a limit 5.191 C higher, or a bound 5.191 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.523 C against a 21.0 C limit, so it fails by 6.523 C. It would take a limit 6.523 C higher, or a bound 6.523 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 29.481 C against a 21.0 C limit, so it fails by 8.481 C. It would take a limit 8.481 C higher, or a bound 8.481 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.193 C against a 21.0 C limit, so it fails by 9.193 C. It would take a limit 9.193 C higher, or a bound 9.193 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.743 C against a 21.0 C limit, so it fails by 9.743 C. It would take a limit 9.743 C higher, or a bound 9.743 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.679 C against a 21.0 C limit, so it fails by 9.679 C. It would take a limit 9.679 C higher, or a bound 9.679 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 29.959 C against a 21.0 C limit, so it fails by 8.959 C. It would take a limit 8.959 C higher, or a bound 8.959 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 29.411 C against a 21.0 C limit, so it fails by 8.411 C. It would take a limit 8.411 C higher, or a bound 8.411 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 27.977 C against a 21.0 C limit, so it fails by 6.977 C. It would take a limit 6.977 C higher, or a bound 6.977 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 26.536 C against a 21.0 C limit, so it fails by 5.536 C. It would take a limit 5.536 C higher, or a bound 5.536 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.606 C against a 21.0 C limit, so it fails by 3.606 C. It would take a limit 3.606 C higher, or a bound 3.606 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 23.840 C against a 21.0 C limit, so it fails by 2.840 C. It would take a limit 2.840 C higher, or a bound 2.840 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.513 C against a 21.0 C limit, so it fails by 0.513 C. It would take a limit 0.513 C higher, or a bound 0.513 C tighter, to change this hour.

23:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.560 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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